S2S Lab 5 Task Solutions

1 Welcome!

2 Sampling Distribution of $\bar{X}$

What is the mean of the random variable X?

3.5

What is the variance of the random variable X?

2.9167

2.1 Sampling with replacement

2.1.1 Sampling distribution for $\bar{X}$

What is the mean of the sampling distribution for $\bar{X}$?

```
means_w <- unique(samples_w$Mean)
mean_Xbar_w <- sum(means_w*samplingdist_mean_w)
mean_Xbar_w
```

```
## [1] 7/2
```

What is the variance of the sampling distribution for $\bar{X}$?

```
means_w <- unique(samples_w$Mean)
var_Xbar_w <- sum(means_w^2*samplingdist_mean_w)-(mean_Xbar_w)^2
var_Xbar_w
```

```
## [1] 35/24
```

2.1.2 Sampling distribution for S^2

What is the mean of the sampling distribution for S^2?

```
vars_w <- unique(samples_w$Variance)
mean_S2_w <- sum(vars_w*samplingdist_var_w)
mean_S2_w
```

```
## [1] 35/12
```

2.2 Sampling without replacement

2.2.1 Sampling distribution of $\bar{X}$

What is the mean of the sampling distribution for $\bar{X}$, when sampling without replacement?

```
means_wo <- unique(samples_wo$Mean)
mean_Xbar_wo <- sum(means_wo*samplingdist_mean_wo)
mean_Xbar_wo
```

```
## [1] 7/2
```

What is the variance of the sampling distribution for $\bar{X}$, when sampling without replacement?

```
means_wo <- unique(samples_wo$Mean)
var_Xbar_wo <- sum(means_wo^2*samplingdist_mean_wo)-(mean_Xbar_wo)^2
var_Xbar_wo
```

```
## [1] 7/6
```

2.2.2 Sampling distribution of S^2

What is the expected value of the sampling distribution for S^2, when sampling without replacement?

```
vars_wo <- unique(samples_wo$Variance)
mean_S2_wo <- sum(vars_wo*samplingdist_var_wo)
mean_S2_wo
```

```
## [1] 7/2
```

3 Functions

3.1 Probability functions

Choose the correct function and complete the code for the following scenarios.

- You want to construct a 90% confidence interval so need to know the 95th quantile of the standard normal distribution.

```
qnorm(p = 0.95, mean = 0, sd = 1)
```

- How would you find the value of x such that $\mathbb{P}(X \leq x) = 0.45$, where $X \sim N(100, 4^2)$?

```
qnorm(p = 0.45, mean = 100, sd = 4)
```

- You want to know the proportion of the $N(0, 2^2)$ distribution that lies below -2. That is, $\mathbb{P}(X \leq -2)$, where $X \sim N(0, 2^2)$.

```
pnorm(q = -2, mean = 0, sd = 2)
```

3.2 Creating functions